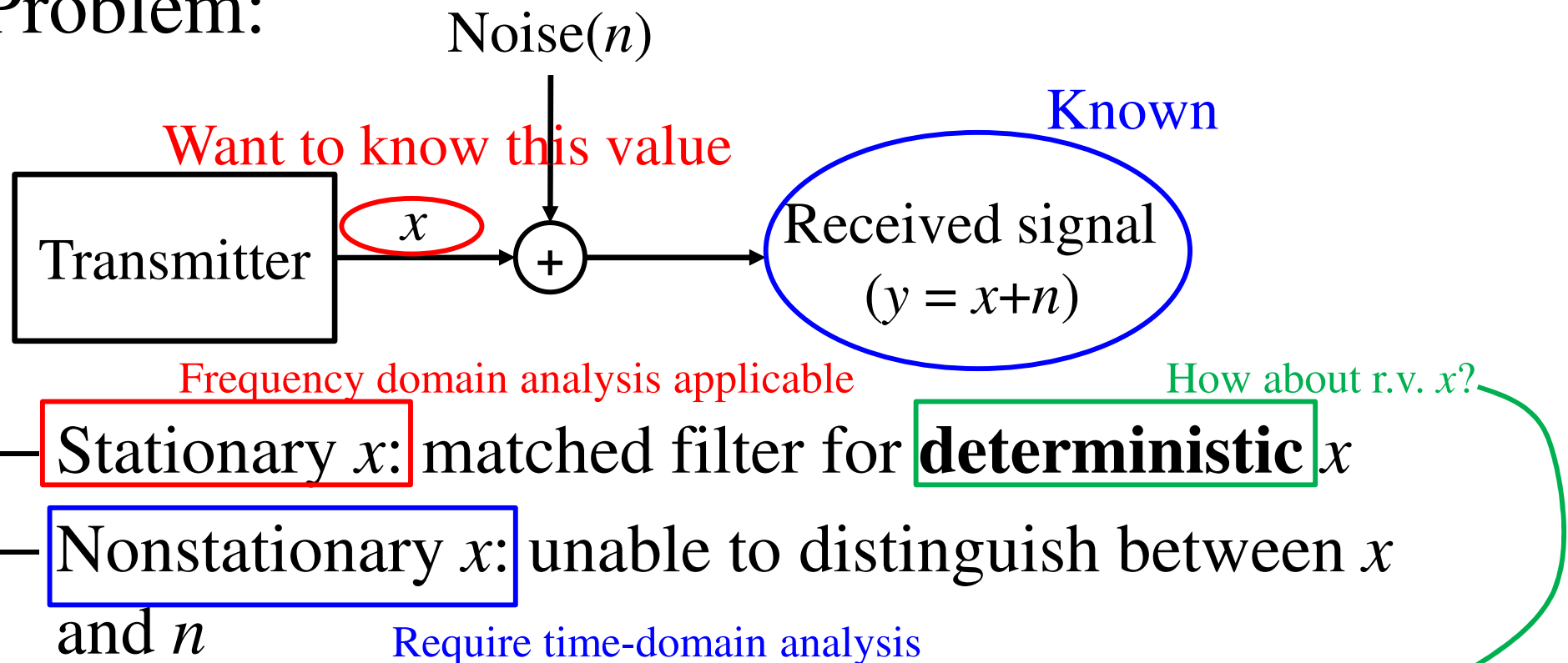


Discrete-Time Kalman Filter

- Construction of Kalman filter

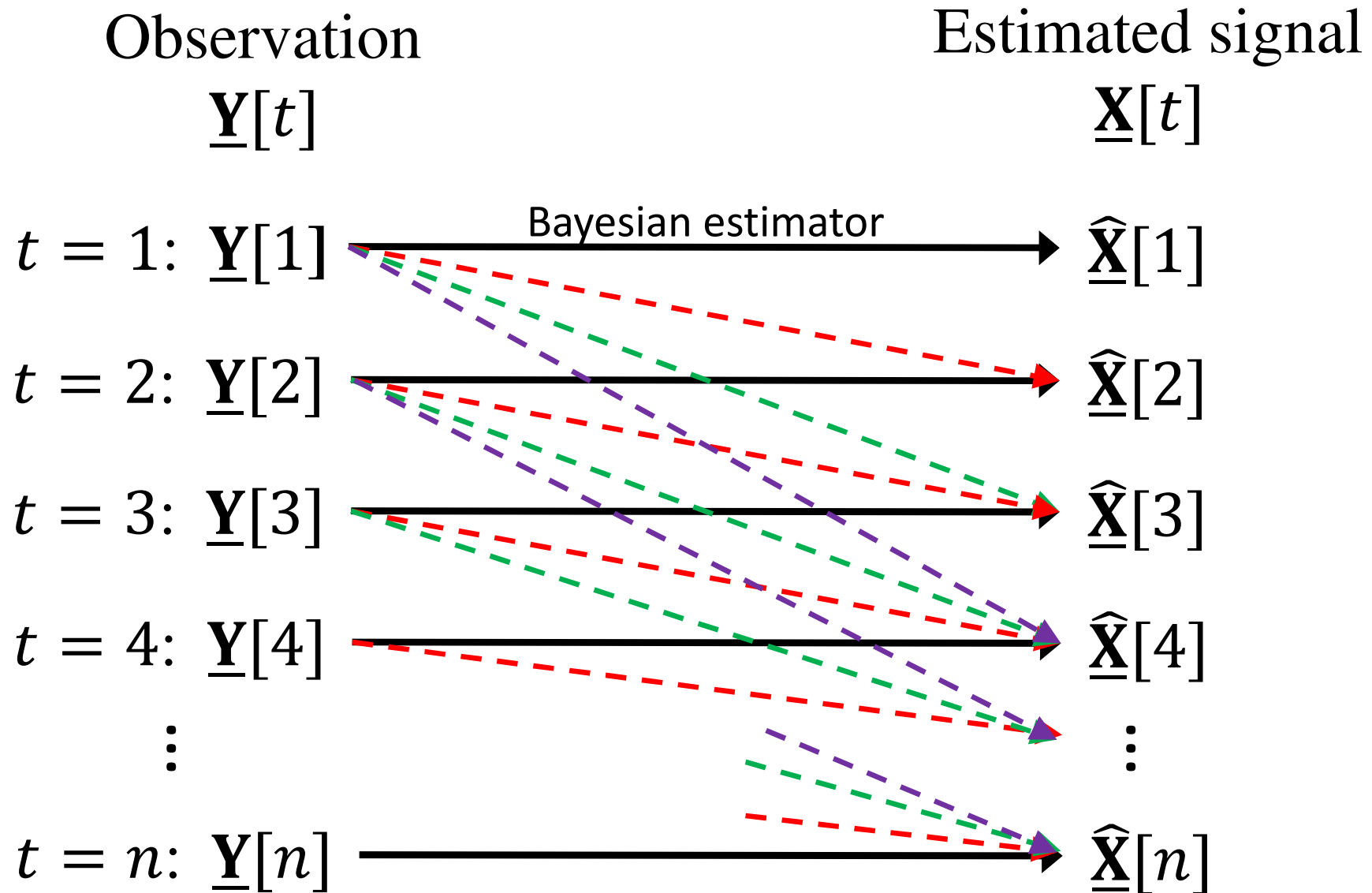
Kalman Filter

- Optimum estimator for conditional mean
- Problem:



Idea: Estimated from y and previously known x .

Limitation of Bayesian Estimator



Unable to use every available information

Kalman Filter: Idea

Signal from sensors

$\underline{\mathbf{Y}}[t]$

Estimated signals

$\underline{\hat{\mathbf{X}}}[t]$

$t = 1: \underline{\mathbf{Y}}[1]$

$t = 2: \underline{\mathbf{Y}}[2]$

$t = 3: \underline{\mathbf{Y}}[3]$

$t = 4: \underline{\mathbf{Y}}[4]$

$\vdots$

$t = n: \underline{\mathbf{Y}}[n]$

$\hat{\underline{\mathbf{X}}}[1]$

$\hat{\underline{\mathbf{X}}}[2]$

$\hat{\underline{\mathbf{X}}}[3]$

$\hat{\underline{\mathbf{X}}}[4]$

$\vdots$

$\hat{\underline{\mathbf{X}}}[n]$

Contain the information of $\underline{\mathbf{Y}}[1], \underline{\mathbf{Y}}[2]$

Contain the information of $\underline{\mathbf{Y}}[1], \underline{\mathbf{Y}}[2], \underline{\mathbf{Y}}[3]$

Contain the information of $\underline{\mathbf{Y}}[1], \underline{\mathbf{Y}}[2], \underline{\mathbf{Y}}[3], \underline{\mathbf{Y}}[4]$.

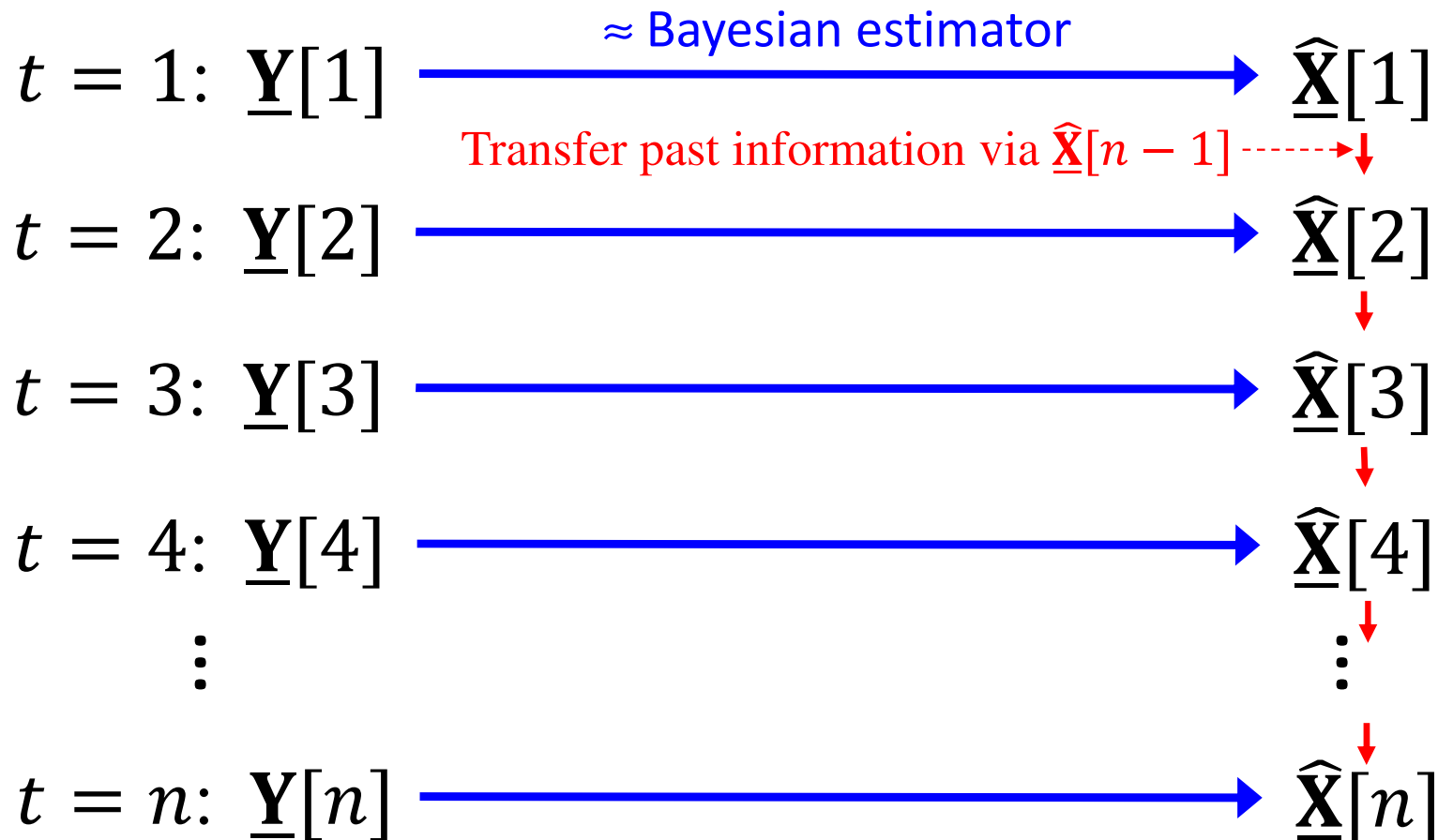
Kalman Filter: Idea (2)

Signal from sensors

$\underline{\mathbf{Y}}[t]$

Estimated signals

$\underline{\mathbf{X}}[t]$



Required Models

Signal from sensors

$\underline{\mathbf{Y}}[n]$

Estimated signals

$\underline{\mathbf{X}}[n]$

Model the change of $\underline{\mathbf{X}}$: State equation

$t = 1: \underline{\mathbf{Y}}[1]$



$\hat{\underline{\mathbf{X}}}[1]$

$t = 2: \underline{\mathbf{Y}}[2]$



$\hat{\underline{\mathbf{X}}}[2]$

$t = 3: \underline{\mathbf{Y}}[3]$



$\hat{\underline{\mathbf{X}}}[3]$

Model relationship between $\underline{\mathbf{X}}$ and $\underline{\mathbf{Y}}$: Observation model

Required models: (1) State Equation
(2) Observation model

Definition

- **State variable**

= “minimum information required for prediction”

- **State equation**

For a continuous-time system, a state equation is usually a linear (or approximately linear) differential equation

= “rules that control (govern) the behavior of the state (system)”

State Equation

$n \times 1$ vector for a state variable $r \times 1$ vector for a disturbance (r.v., noise)

Continuous time: $\dot{\underline{\mathbf{X}}}(t) = \underline{\mathbf{A}}(t)\underline{\mathbf{X}}(t) + \underline{\mathbf{B}}(t)\underline{\mathbf{V}}(t)$

$$\dot{\underline{\mathbf{X}}}(t) = \frac{d\underline{\mathbf{X}}(t)}{dt}$$

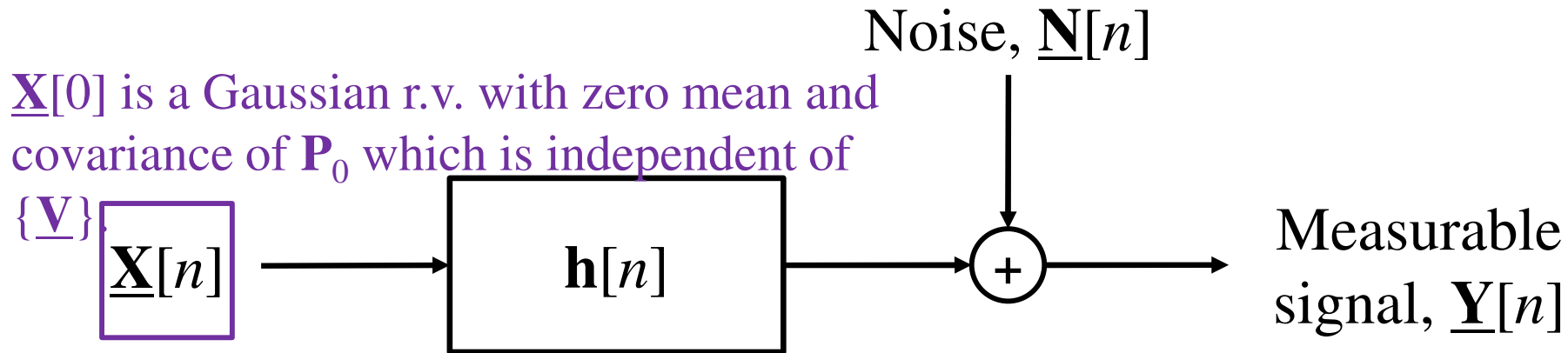
$\underline{\mathbf{A}} \in R^{n \times n}$, $\underline{\mathbf{B}} \in R^{n \times r}$ indicating the system's behavior (deterministic)

Discrete time: $\underline{\mathbf{X}}_{n+1} = \underline{\mathbf{A}}[n]\underline{\mathbf{X}}_n + \underline{\mathbf{B}}[n]\underline{\mathbf{V}}_n$

$\underline{\mathbf{X}}$ in next state

Use difference equation instead of differential equation.

Kalman Filter: Models



For simplicity, consider zero mean white Gaussian noise with $E[\underline{\mathbf{V}}[n]\underline{\mathbf{V}}^T[n] = \mathbf{Q}(n)$

State equation: $\dot{\underline{\mathbf{X}}}[n] = \underline{\mathbf{A}}[n]\underline{\mathbf{X}}[n] + \underline{\mathbf{B}}[n]\underline{\mathbf{V}}[n]$

Known

Observation model: $\underline{\mathbf{Y}}[n] = \underline{\mathbf{h}}[n]\underline{\mathbf{X}}[n] + \underline{\mathbf{N}}[n]$

White Gaussian measurement noise independent of $\underline{\mathbf{X}}[n]$ and $\underline{\mathbf{V}}[n]$ with zero mean and *nonstationary* covariance $\mathbf{R}[n]$.

Simplified Version

- Scalar $Y[n]$ (measure one value at a time)
- Discrete time system

$$\underline{\mathbf{X}}[n + 1] = \mathbf{A}[n]\underline{\mathbf{X}}[n] + \mathbf{B}[n]\underline{\mathbf{V}}[n]$$

$$Y[n] = \mathbf{h}^T[n]\underline{\mathbf{X}}[n] + N[n]$$

Required information

Nonstationary covariance of $Y[n]$

Time-varying deterministic parameters: $\mathbf{A}[n], \mathbf{B}[n], \mathbf{h}[n]$

Covariance of $\underline{\mathbf{X}}[0]$

Stochastic parameters: $\mathbf{P}[0] = \mathbf{P}_0, \mathbf{Q}[n], R[n]$

Observable output: $Y[n]$

Nonstationary covariance of $\underline{\mathbf{V}}(t)$

Kalman Filter: Initial Summary

- Required model: System (state) knowledge + noise

$$\underline{\mathbf{X}}[n] = \mathbf{A}[n-1]\underline{\mathbf{X}}[n-1] + \mathbf{B}[n-1]\underline{\mathbf{V}}[n-1]$$

$$Y[n] = \mathbf{h}^T[n]\underline{\mathbf{X}}[n] + N[n]$$

Observation (measurement) model + noise

- Output: MMSE estimation of $\underline{\mathbf{X}}(n)$

$$\underline{\mathbf{Z}}[n] = [Y[1] \ Y[2] \ \dots \ Y[n]]$$

$$\begin{aligned}\hat{\underline{\mathbf{X}}}[n] &= E(\underline{\mathbf{X}}[n] | Y[n], Y[n-1], \dots Y[2], Y[1]) \\ &= E(\underline{\mathbf{X}}[n] | \underline{\mathbf{Z}}[n])\end{aligned}$$

Kalman Filter: Propagate Formula

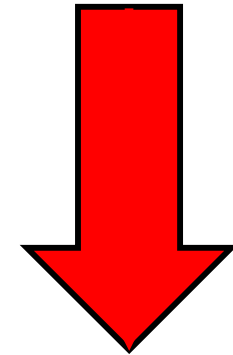
1) Propagate past information (via state equation) to the prediction of the current $\underline{\mathbf{X}}[n]$.

$$\underline{\mathbf{X}}[n] = \mathbf{A}[n-1]\underline{\mathbf{X}}[n-1] + \mathbf{B}[n-1]\underline{\mathbf{V}}[n-1]$$

$$\underline{\mathbf{X}}_n = \mathbf{A}_{n-1}\underline{\mathbf{X}}_{n-1} + \mathbf{B}_{n-1}\underline{\mathbf{V}}_{n-1}$$

$$y[n] \xrightarrow{\hspace{10em}} \hat{\underline{\mathbf{X}}}[n]$$

$$\hat{\underline{\mathbf{X}}}[n-1]$$



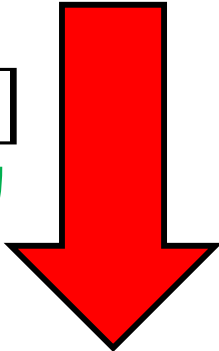
Mean provides the estimation with MMSE so the prediction of $\underline{\mathbf{X}}_n$ is

$$\begin{aligned} E(\underline{\mathbf{X}}_n | \underline{\mathbf{Z}}_{n-1}) &= E(\underbrace{\mathbf{A}_{n-1}\underline{\mathbf{X}}_{n-1}}_{\text{Deterministic}} + \underbrace{\mathbf{B}_{n-1}\underline{\mathbf{V}}_{n-1}}_{\text{Independent of } y[\cdot] \text{ and zero mean}} | \underline{\mathbf{Z}}_{n-1}) \\ &= \mathbf{A}_{n-1} \underbrace{E(\underline{\mathbf{X}}_{n-1} | \underline{\mathbf{Z}}_{n-1})}_{\text{Estimation of } \underline{\mathbf{X}}[n-1] \text{ given the past information.}} + \mathbf{0} = \mathbf{A}_{n-1} \hat{\underline{\mathbf{X}}}_{n-1} \end{aligned}$$

Kalman Filter: Prediction Error

$$\underline{\mathbf{X}}[n] = \underbrace{\mathbf{A}[n-1]\underline{\mathbf{X}}[n-1]}_{\substack{\text{Deterministic} \\ \text{Estimation error of } \hat{\underline{\mathbf{X}}}[n-1]: \\ \mathbf{P}[n-1]}} + \underbrace{\mathbf{B}[n-1]\underline{\mathbf{V}}[n-1]}_{\substack{\text{Deterministic} \\ \text{Noise}}}$$

$\hat{\underline{\mathbf{X}}}[n-1]$



$y[n] \xrightarrow{\hspace{10em}} \hat{\underline{\mathbf{X}}}[n]$

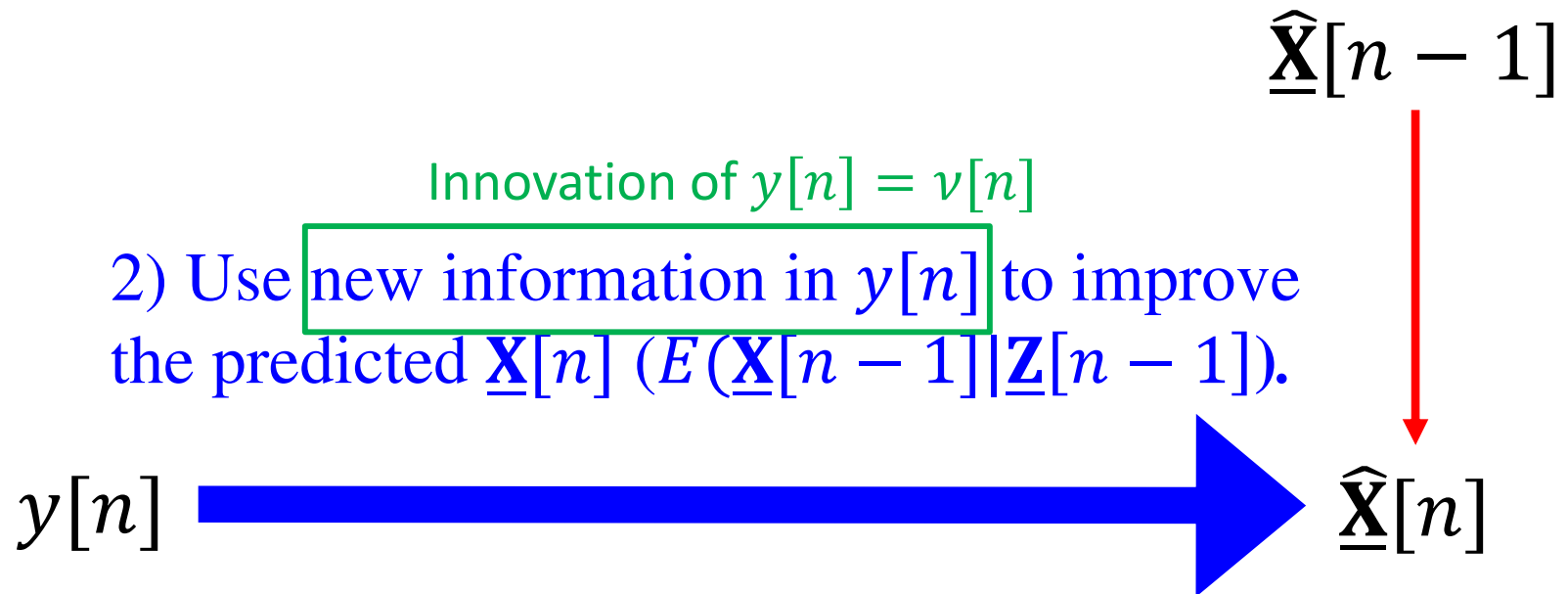
Covariance is related to the MSE when we estimate $\hat{\underline{\mathbf{X}}}$ by its mean.

$$\mathbf{M}[n] = \mathbf{A}[n-1]\mathbf{P}[n-1]\mathbf{A}^T[n-1] + \mathbf{B}[n-1]\mathbf{Q}[n-1]\mathbf{B}^T[n-1],$$

For $\underline{\mathbf{Y}} = \mathbf{A}\underline{\mathbf{X}}$, we have $\mathbf{C}_{\underline{\mathbf{Y}}\underline{\mathbf{Y}}} = \mathbf{A}\mathbf{C}_{\underline{\mathbf{X}}\underline{\mathbf{X}}}\mathbf{A}^T$

where $\mathbf{M}[n]$ is the prediction error covariance matrix.

Kalman Filter: Update Formula



Bayesian estimator when the world is after $\underline{\mathbf{Z}}[n-1]$ has occurred.

$$\hat{\underline{\mathbf{X}}}[n] = E(\hat{\underline{\mathbf{X}}}[n]|\underline{\mathbf{Z}}[n-1]) + \mathbf{C}_{\hat{\underline{\mathbf{X}}}_n v_n} C_{v_n v_n}^{-1} v_n$$

Formulate iterative function to evaluate these two values.

$$= \mathbf{A}[n-1]\hat{\underline{\mathbf{X}}}[n-1] + \underline{\alpha}_{nn} d_n^{-1} v_n$$

Innovation

$$v_n = y[n] - E(y[n] | \underline{\mathbf{Z}}[n-1])$$

$$= y[n] - E(y[n] | \hat{\underline{\mathbf{X}}}[n-1])$$

$$= y[n] - E(\overset{\text{Deterministic}}{\boxed{h^T[n]}} \underline{\mathbf{X}}[n] + \overset{\text{Independent of } \hat{\underline{\mathbf{X}}}[\cdot] \text{ and zero mean}}{\boxed{N[n]}} | \hat{\underline{\mathbf{X}}}[n-1])$$

$$= y[n] - h^T[n] \overset{\text{Prediction of } \underline{\mathbf{X}}[n]}{\boxed{E(\underline{\mathbf{X}}[n] | \hat{\underline{\mathbf{X}}}[n-1])}} + 0$$

$$= y[n] - h^T[n] \mathbf{A}[n-1] \hat{\underline{\mathbf{X}}}[n-1]$$

Kalman Filter: Estimation Error

MSE of Bayesian estimator = $tr(\mathbf{P})$ The reduction of MSE of $\underline{\mathbf{X}}$ by $\underline{\mathbf{Y}}$

Gaussian random variables: $\mathbf{P} = \mathbf{C}_{\underline{\mathbf{X}}\underline{\mathbf{X}}} - \mathbf{C}_{\underline{\mathbf{X}}\underline{\mathbf{Y}}} \mathbf{C}_{\underline{\mathbf{Y}}\underline{\mathbf{Y}}}^{-1} \mathbf{C}_{\underline{\mathbf{Y}}\underline{\mathbf{X}}}$

In the update formula, the prediction is made more accurate by an innovation.

$$\mathbf{P}[n] = \underbrace{\mathbf{M}[n]}_{\text{Prediction error}} - \underbrace{\underline{\alpha}_{nn} d_n^{-1} \underline{\alpha}_{nn}^T}_{\text{Improvement by the addition of an innovation}}$$

Formulation for $\underline{\alpha}_{nn}$ and d_n

From the direct evaluation.

$$\underline{\alpha}_{nn} = E(\underline{\mathbf{X}}_n v_n) - E(\underline{\mathbf{X}}_n)E(v_n)$$

$$\underline{\alpha}_{nn} = \mathbf{M}[n]\mathbf{h}[n]$$

$$d_n = E(v_n v_n) - E^2(v_n)$$

$$d_n = \mathbf{h}^T[n]\mathbf{M}[n]\mathbf{h}[n] + R[n]$$

Errors in Kalman Filter

- Causes: Randomness in the form of noise ($\underline{\mathbf{V}}(t)$ and $N(t)$)

- Prediction error: (error from predictor)

$$\underline{\Delta}(t) = \underline{\mathbf{X}}(t) - E[\underline{\mathbf{X}}(t) | \underline{\zeta}(t-1)]$$

Error covariance matrix (smaller is better)

$$\boxed{\mathbf{M}(t)} = E[\underline{\Delta}(t) \underline{\Delta}^T(t)]$$

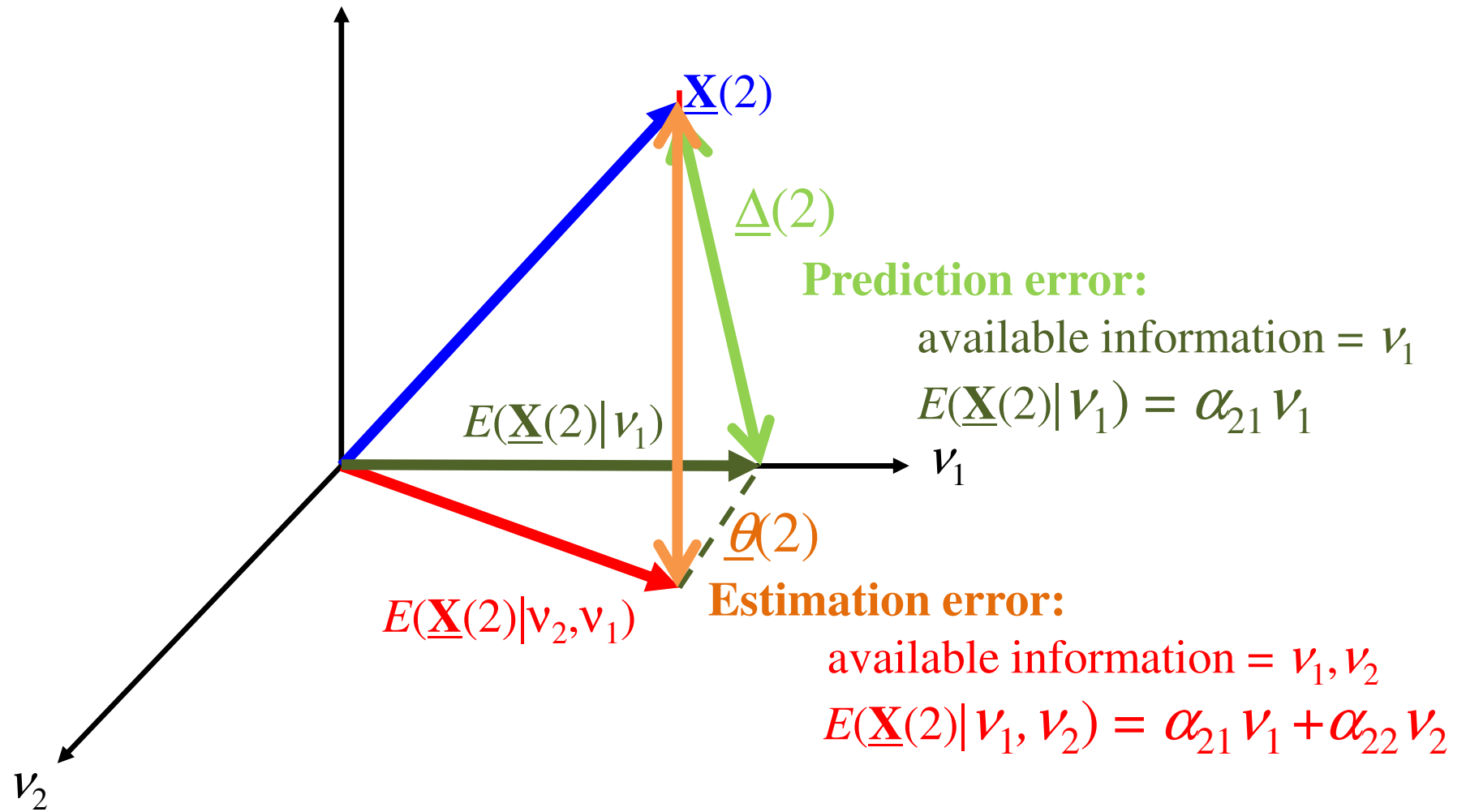
- Estimation error: Update based on innovation.

$$\underline{\theta}(t) = \underline{\mathbf{X}}(t) - E[\underline{\mathbf{X}}(t) | \underline{\zeta}(t)] = \underline{\mathbf{X}}(t) - \boxed{\hat{\underline{\mathbf{X}}}(t)}$$

$$\mathbf{P}(t) = E[\underline{\theta}(t) \underline{\theta}^T(t)]$$

Error: Illustration

At $t = 2$:



Statistics of Innovation: Calculation

- Mean: $\mu_v = E[v(t)] = 0$
- Variance: $d_t = E[v^2(t)] - E^2[v(t)] = E[v^2(t)]$

$$= E\left\{\left[\underbrace{\mathbf{h}^T(t)\underline{\Delta}(t)}_{\text{Deterministic}} + N(t)\right]\left[\underbrace{\underline{\Delta}^T(t)\mathbf{h}(t)}_{E(\bullet)=0} + N(t)\right]\right\}$$

$$= E\left[\underbrace{\mathbf{h}^T(t)\underline{\Delta}(t)\underline{\Delta}^T(t)\mathbf{h}(t)}_{\text{Prediction error covariance matrix, } \mathbf{M}(t)} + 2\underbrace{N(t)\underline{\Delta}^T(t)\mathbf{h}(t)}_{\text{Covariance of } N, R(t)} + N^2(t)\right]$$

$$= \mathbf{h}^T(t)E\left[\underline{\Delta}(t)\underline{\Delta}^T(t)\right]\mathbf{h}(t) + E\left[N^2(t)\right]$$

$$d_t = \mathbf{h}^T(t)\mathbf{M}(t)\mathbf{h}(t) + R(t)$$

Statistics of Innovation: Calculation (2)

- Cross-covariance (projection to $\mathbf{v}_k$ axis)

$$\begin{aligned}
 \underline{\alpha}_{tk} &= E[\underline{\mathbf{X}}(t)\mathbf{v}(k)] \\
 &= E[\underline{\mathbf{X}}(t)\{\Delta^T(k)\mathbf{h}(k) + N(k)\}] \\
 &\quad \text{Deterministic} \\
 &= E[\underline{\mathbf{X}}(t)\Delta^T(k)]\mathbf{h}(k) + E[\underline{\mathbf{X}}(t)N(k)] \\
 &\quad \Delta(t) = \underline{\mathbf{X}}(t) - E[\underline{\mathbf{X}}(t)|\zeta(t-1)] \quad 0 \\
 &\quad \text{Deterministic} \\
 &= E[\{\Delta(t) + E[\underline{\mathbf{X}}(t)|\zeta(t-1)]\}\Delta^T(k)]\mathbf{h}(k) \\
 &\quad \mathbf{M}(t) \quad 0 \\
 &= E[\Delta(t)\Delta^T(k)] + E[\underline{\mathbf{X}}(t)|\zeta(t-1)]E[\Delta^T(t)]\mathbf{h}(k) \\
 &\quad 0 \\
 \underline{\alpha}_{tk} &= \mathbf{M}(t)\mathbf{h}(k)
 \end{aligned}$$

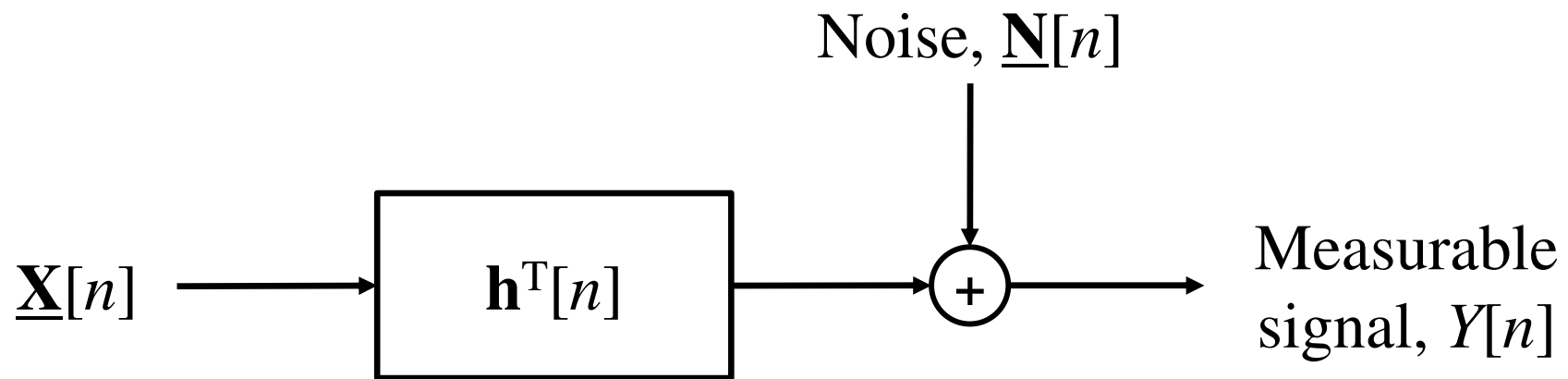
Kalman Filter: Models

$$\underline{\mathbf{X}}[n + 1] = \mathbf{A}[n]\underline{\mathbf{X}}[n] + \mathbf{B}[n]\underline{\mathbf{V}}[n]$$

Deterministic. Assume known.

$$Y[n] = \mathbf{h}^T[n]\underline{\mathbf{X}}[n] + N[n]$$

Assume noise statistics are known.



Kalman Filter: Equations

Kalman Filter Equations:

Kalman gain: $\mathbf{k}_n$

$$\hat{\underline{\mathbf{X}}}[n] = \mathbf{A}[n-1]\hat{\underline{\mathbf{X}}}[n-1] + \frac{\underline{\alpha}_{nn}}{d_n}v_n$$

Innovation and its statistic:

$$v_n = Y[n] - \mathbf{h}^T[n]\mathbf{A}[n-1]\hat{\underline{\mathbf{X}}}[n-1]$$

$$\underline{\alpha}_{nn} = \mathbf{M}[n]\mathbf{h}[n]$$

$$d_n = \mathbf{h}^T[n]\mathbf{M}[n]\mathbf{h}[n] + R[n]$$

Prediction error covariance:

$$\mathbf{M}_n = \mathbf{A}_{n-1}\mathbf{P}_{n-1}\mathbf{A}_{n-1}^T + \mathbf{B}_{n-1}\mathbf{Q}_{n-1}\mathbf{B}_{n-1}^T$$

Estimation error covariance: $\mathbf{P}_n = \mathbf{M}_n - \underline{\alpha}_{nn}d_n^{-1}\underline{\alpha}_{nn}^T$